

PROJECT DECISION LOG

Introduction

This Decision Log records the significant business, technical, and design decisions made throughout the HPO Cost Efficiency Analytics Solution, an enterprise analytics solution developed for the Automotive Manufacturing industry with a focus on Cost Efficiency and Operational Excellence. Maintaining a structured record of project decisions supports governance, transparency, traceability, knowledge transfer, and future enhancements. It complements the project's journal, version history, and technical documentation by providing the rationale behind key implementation choices.

Decision 001

Decision

Develop the solution as an enterprise analytics platform rather than a practice dashboard.

Reason

To demonstrate Finance Planning, Business Analysis, and Business Intelligence capabilities through a realistic business solution.

Decision 002

Decision

Adopt a phased development approach.

Reason

Improves planning, progress tracking, version control, and scope management.

Decision 003

Decision

Use a Star Schema data model with separate Fact and Dimension tables.

Reason

Supports scalability, simplifies reporting, and aligns with Power BI best practices.

Decision 004

Decision

Create Dimension tables using Reference queries.

Reason

Maintains linkage to source queries and simplifies future maintenance.

Decision 005

Decision

Move business logic from Excel into Power BI.

Reason

Creates a self-contained semantic model and demonstrates DAX capabilities.

Decision 006

Decision

Use Calculated Columns for row-level business calculations.

Reason

Row-level metrics such as scores and payback belong to individual initiatives.

Decision 007

Decision

Use Measures for aggregated KPIs.

Reason

Measures provide dynamic calculations based on report filters and interactions.

Decision 008

Decision

Implement a weighted business scoring framework.

Reason

Prioritization combines savings, payback, ease, strategic alignment, investment, and business assumptions.

Decision 009

Decision

Base scoring assumptions on business priorities.

Reason

Reflects realistic management decision-making rather than arbitrary technical values.

Decision 010

Decision

Design dashboards around stakeholder requirements.

Reason

Each report page addresses the needs of its intended audience.

Decision 011

Decision

Maintain GitHub documentation continuously.

Reason

Provides transparent development history and project governance.

Decision 012

Decision

Complete one dashboard page before starting the next.

Reason

Ensures validation, documentation, and controlled delivery.

Decision 013

Decision

Maintain separate Version 1 and Version 2 backlogs.

Reason

Delivers essential functionality first while controlling scope.

Decision 014

Decision

Keep the solution generic rather than company-specific.

Reason

Improves portfolio relevance across industries.

Decision 015**Decision**

Design the Initiative Tracker as an operational dashboard.

Reason

Supports day-to-day monitoring through operational reporting.

Decision 016**Decision**

Use dropdown slicers across dashboards.

Reason

Optimizes screen space while preserving usability.

Decision 017**Decision**

Combine summary visuals with detailed initiative tables.

Reason

Enables drill-down from summaries to transaction-level information.

Decision 018**Decision**

Prioritize dashboard readability over visual density.

Reason

Improves usability and executive consumption.

Decision 019**Decision**

Design the Financial Impact dashboard for FP&A; users.

Reason

Supports forecasting, budgeting, and financial planning.

Decision 020**Decision**

Use forecast-based financial visualizations and compare Savings versus Implementation Cost. **Reason**

Improves investment evaluation and planning visibility.

Decision 021**Decision**

Use Net Savings as the primary financial performance metric.

Reason

Represents value after implementation costs.

Decision 022**Decision**

Defer final formatting until all pages are complete.

Reason

Ensures consistent presentation across the solution.

Decision 023

Decision

Create a dedicated Executive Recommendation dashboard.

Reason

Provides focused executive decision support.

Decision 024

Decision

Generate workflow-aware Recommended Actions.

Reason

Recommendations align with initiative lifecycle and investment score.

Decision 025

Decision

Normalize strategic scoring using a 1–5 scale.

Reason

Improves consistency and comparability.

Decision 026

Decision

Display financial values in Lakhs without currency symbols.

Reason

Improves readability and reporting consistency.

Conclusion

Maintaining this Decision Log establishes a permanent record of the governance and evolution of this Automotive Manufacturing Cost Efficiency and Operational Excellence analytics solution. It supports future maintenance, knowledge transfer, continuous improvement, and long-term scalability by preserving the rationale behind significant implementation decisions.